

Section A

For each question there are four possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

- 1** Which one of the following statements about one mole of metal is always true?
- A** It contains the same number of particles as 24 dm³ of hydrogen gas.
- B** It contains the same number of atoms as $\frac{1}{12}$ mole of carbon-12 isotope.
- C** It has the same mass as one mole of hydrogen atoms.
- D** It is liberated by one mole of electrons.

- 2** The process 'efflorescence' takes place when washing powder containing hydrated sodium carbonate with the formula Na₂CO₃·10H₂O, loses some of its water of crystallisation on exposure to the atmosphere.

In one experiment, 3.17 g of washing powder was left in the open. After some time, the mass of the powder decreased to 2.17 g.

Given that the formula of effloresced washing powder is Na₂CO₃·xH₂O, What is the value of x?

- A** 2 **B** 3 **C** 4 **D** 5
- 3** Naturally occurring iron consists of four isotopes, ⁵⁴Fe, ⁵⁶Fe, ⁵⁷Fe and ⁵⁸Fe. Two ions formed by the isotopes have chemical formulae shown below:



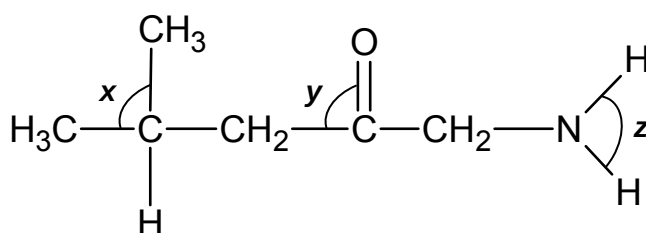
Which one of the following statements is true?

- A** The electron configuration of both Fe²⁺ ions is 1s²2s²2p⁶3s²3p⁶3d⁶4s².
- B** When subjected to an electric field, ⁵⁶/₂₆Fe²⁺ ion gives a greater angle of deflection than ⁵⁴/₂₆Fe²⁺ ion.
- C** ⁵⁶/₂₆Fe²⁺ ion has more protons in its nucleus than ⁵⁴/₂₆Fe²⁺ ion.
- D** The total number of protons and electrons in each of the Fe²⁺ ions is smaller than the nucleon number of the respective ion.

4 Which one of the following pairs contains two species with different shapes?

- A NCl_3 and BrCl_3
 B CO_3^{2-} and BBr_3
 C BeCl_2 and BrCl_2^-
 D SiF_4 and PF_4^+

5 What are the bond angles x , y and z in the molecule below?



- | | x | y | z |
|---|-------------|-------------|-------------|
| A | 90° | 109° | 120° |
| B | 107° | 120° | 109° |
| C | 109° | 120° | 107° |
| D | 120° | 90° | 109° |

6 Silicon carbide, SiC , is a major industrial abrasive and a refractory material which melts at a very high temperature.

Which one of the following structures exists in SiC that helps to explain the above properties?

- A A giant ionic lattice structure with ionic bonds between silicon and carbon ions.
 B A giant molecular structure with strong covalent bonds between silicon and carbon atoms forming a 3 dimensional network.
 C A layered structure with covalent bonds between silicon and carbon atoms and weak van der Waals' between the layers.
 D A simple molecular structure with covalent bonds between silicon and carbon atoms and weak van der Waals' between SiC molecules.

- 7 Which one of the following pairs of substances has different types of bonding and structure?

A	AlF_3	AlCl_3	B	MgF_2	BaI_2
C	Diamond	SiO_2	D	PCl_3	SiCl_3H

- 8 When 50 cm^3 of $1 \text{ mol dm}^{-3} \text{ NaOH(aq)}$ was added to 10 cm^3 of $1 \text{ mol dm}^{-3} \text{ H}_2\text{SO}_4\text{(aq)}$, a temperature rise of 5°C was obtained.

Assuming that there is negligible heat loss to the surroundings, what is the enthalpy change of neutralization?

(Density of solution may be taken as 1 g cm^{-3} and specific heat capacity of the solution as $4.2 \text{ J g}^{-1}\text{K}^{-1}$)

A	$-\frac{60 \times 4.2 \times 5 \times 1000}{10} \text{ kJ mol}^{-1}$	B	$-\frac{60 \times 4.2 \times 5}{10} \text{ kJ mol}^{-1}$
C	$-\frac{60 \times 4.2 \times 5 \times 1000}{20} \text{ kJ mol}^{-1}$	D	$-\frac{60 \times 4.2 \times 5}{20} \text{ kJ mol}^{-1}$

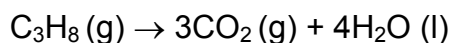
- 9 Given the following data:

$$\Delta H_c \text{ of C(s)} = -393 \text{ kJ mol}^{-1}$$

$$\Delta H_f \text{ of H}_2\text{O(l)} = -285 \text{ kJ mol}^{-1}$$

$$\Delta H_f \text{ of C}_3\text{H}_8\text{(g)} = -99 \text{ kJ mol}^{-1}$$

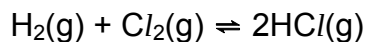
Calculate the enthalpy change for the following reaction.



- A** -579 kJ mol^{-1}
B $+579 \text{ kJ mol}^{-1}$
C $-2220 \text{ kJ mol}^{-1}$
D $+2220 \text{ kJ mol}^{-1}$

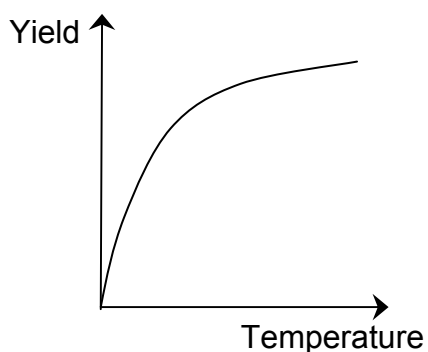
- 10 Use of the Data Booklet is relevant to this question.

Hydrogen chloride can be synthesized directly using a mixture of hydrogen gas and chlorine gas in a reversible reaction as shown:

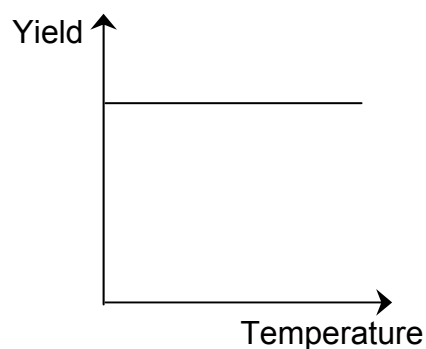


Which one of the following describes the yield of hydrogen chloride at equilibrium as temperature increases?

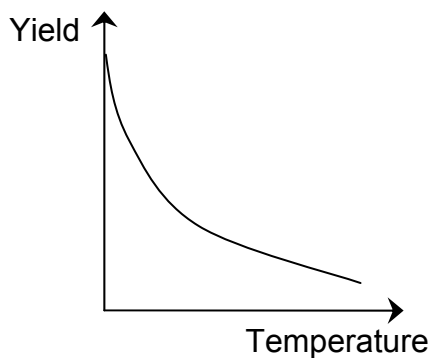
A



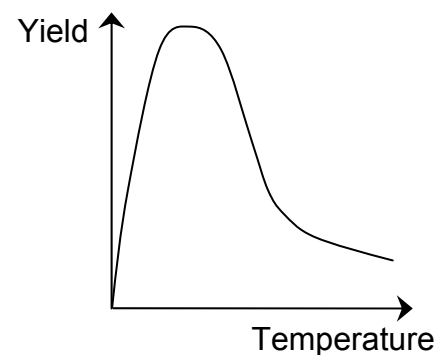
B



C



D

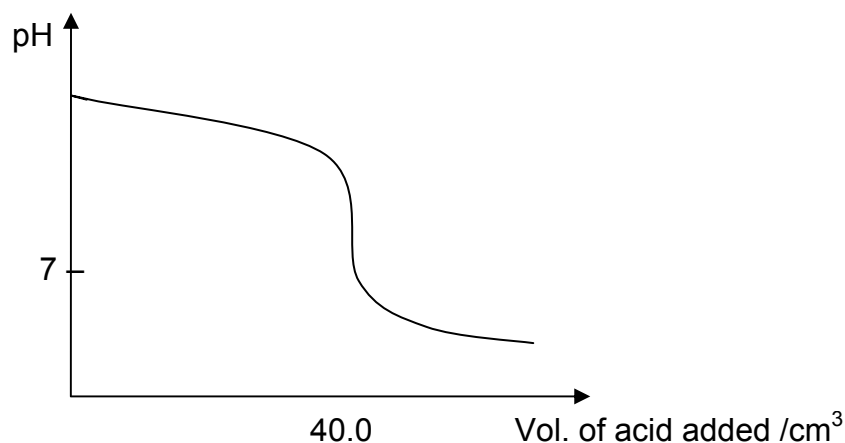


- 11 What is the final pH of a solution formed by mixing 20 cm³ of solution **A** at pH 3.0 and 50 cm³ of solution **B** at pH 5.0?

A 1.7 **B** 3.0 **C** 3.3 **D** 3.5

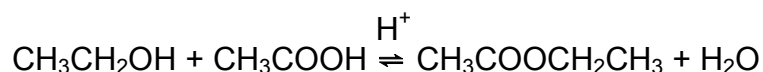
- 12 The following graph shows the changes in pH of 20.0 cm³ of 1.0 mol dm⁻³ alkali solution when excess 1.0 mol dm⁻³ acid solution is added gradually.

Which one of the following pairs of solution with a suitable indicator could have resulted in the graph below?



	Alkali	Acid	Indicator
A	Ba(OH) ₂	HCl	Phenolphthalein
B	Ba(OH) ₂	CH ₃ CO ₂ H	Phenolphthalein
C	NaOH	H ₂ SO ₄	Methyl orange
D	NaOH	CH ₃ CO ₂ H	Methyl orange

- 13 Ethyl ethanoate is formed by the reaction between ethanol and ethanoic acid as shown below:



Given that the above acid-catalyzed esterification is first order with respect to the hydrogen ion, and assuming that all other conditions are kept constant, what is the ratio $t_{\text{pH } 3} : t_{\text{pH } 1}$?

($t_{\text{pH } n}$ = time taken for the reaction conducted at pH = n to achieve 80% yield)

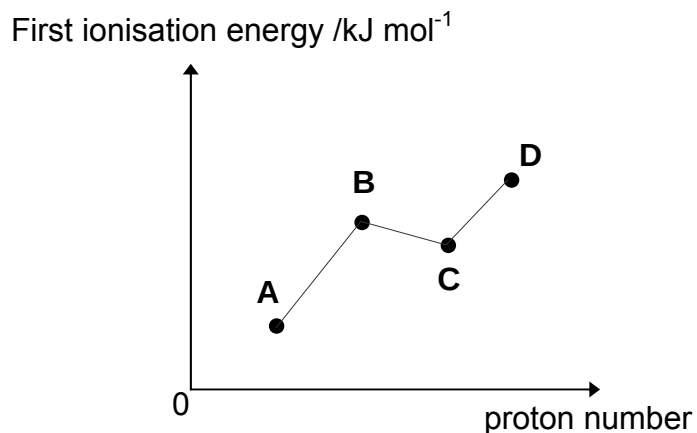
- A 0.01 B 0.33 C 3 D 100

- 14 Radioactive decay obeys first order kinetics. The amount of radioactivity recorded from an isotope decreased from 2720 units to 85 units in 45 hours.

What is the time required for the radioactivity of the same isotope to drop from 2400 units to 300 units?

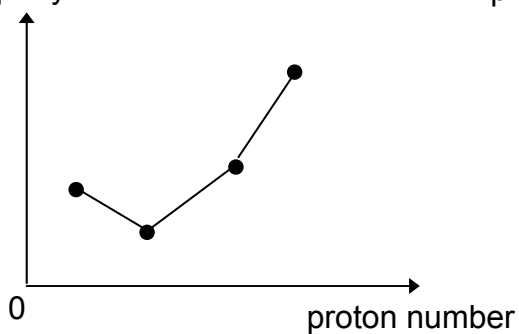
- A 9 hours B 18 hours C 27 hours D 36 hours

- 15 The diagram represents the first ionisation energy of four consecutive elements in the third period of the Periodic Table, with element **B** showing two oxidation states in its chlorides.

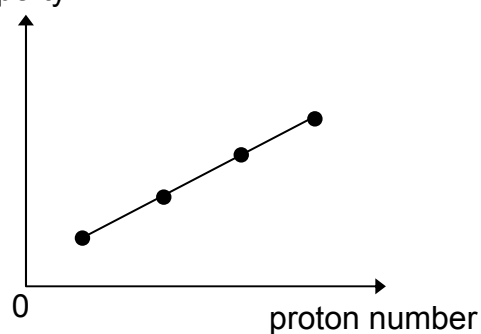


The sketches below represent another two properties of the elements.

property **M**



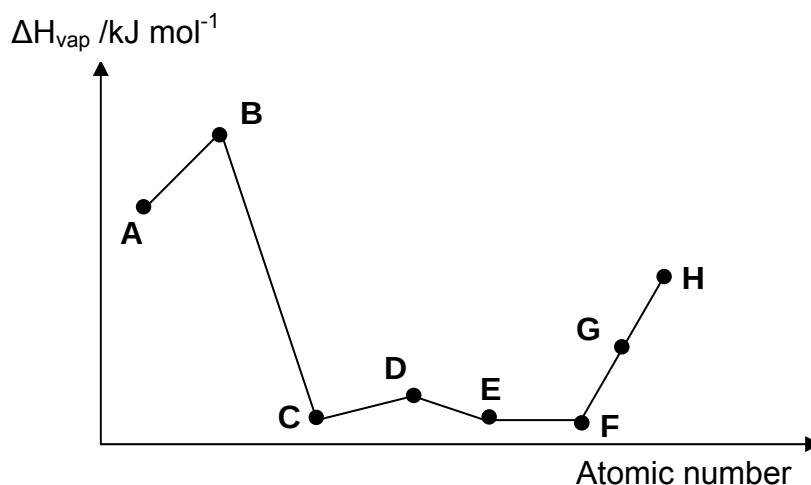
property **N**



What are properties **M** and **N**?

	property M	property N
A	third ionisation energy	electronegativity
B	number of valence electrons	boiling point
C	ionic radius	effective nuclear charge
D	electrical conductivity	atomic radius

- 16 The graph below shows the variation in the enthalpy change of vaporization, ΔH_{vap} for 8 consecutive elements in the Periodic Table, all with atomic number ≤ 20 .



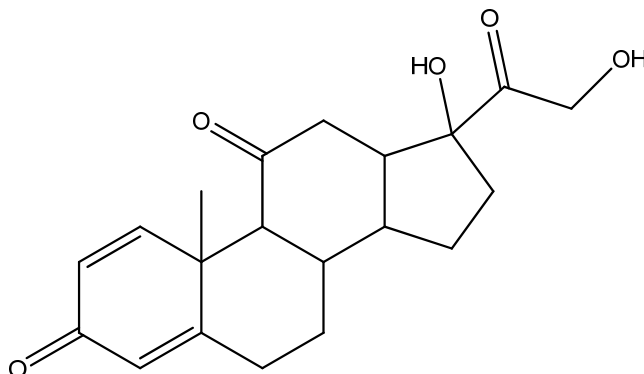
What can be deduced from the above graph?

- A E is soluble in warm benzene whereas F is not.
- B The oxide of A reacts with excess aqueous sodium hydroxide to form a soluble complex.
- C The pH of a solution containing a mixture of oxide of G and oxide of D is greater than 7.
- D The chlorides become less acidic from A to C.
- 17 Compounds of Period 3 elements dissolve in water to form aqueous solutions that are acidic, basic or neutral.

Which one of the following sequences shows the order of **decreasing** pH of the resultant solutions formed when the compounds are dissolved in water?

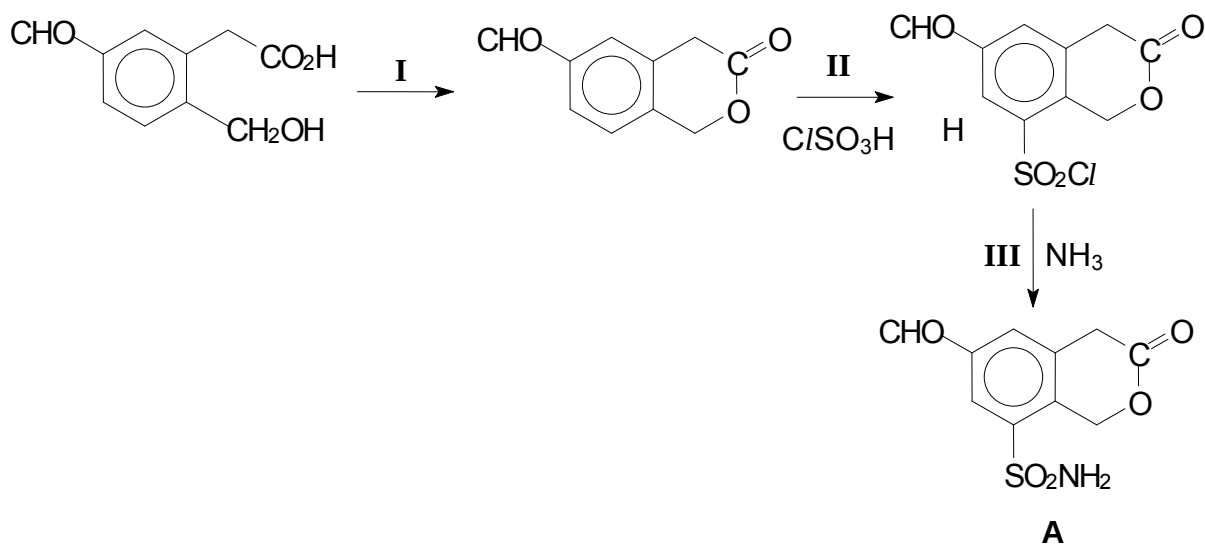
- A MgO , AlCl_3 , P_4O_{10}
- B SiCl_4 , MgCl_2 , NaCl
- C PCl_5 , SiCl_4 , AlCl_3
- D SO_2 , MgO , Al_2O_3

- 18 Prednisone is an immune suppressant used to prevent rejection after organ transplants.



Which functional group is **not** found in the molecule?

- A Ester
 - B Alkene
 - C Alcohol
 - D Ketone
- 19 Compound **A** is synthesized in a three-step synthesis below.



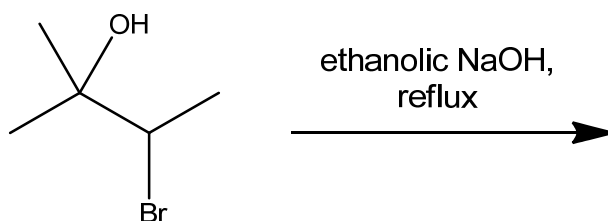
What type of reaction do steps **I** to **III** each illustrate?

- | | I | II | III |
|----------|--------------|--------------|--------------|
| A | Substitution | Substitution | Substitution |
| B | Substitution | Addition | Substitution |
| C | Condensation | Substitution | Substitution |
| D | Condensation | Addition | Substitution |

20 In which one of the following processes is at least one product a gas at room temperature and pressure?

- A oxidation of ethanal by acidified potassium dichromate(VI)
- B dehydration of ethanol
- C esterification of ethanoic acid by ethanol
- D substitution of ethanol by hydrogen bromide

21 Compound **W** is the major product of the following reaction.



What is the structure of compound **W**?

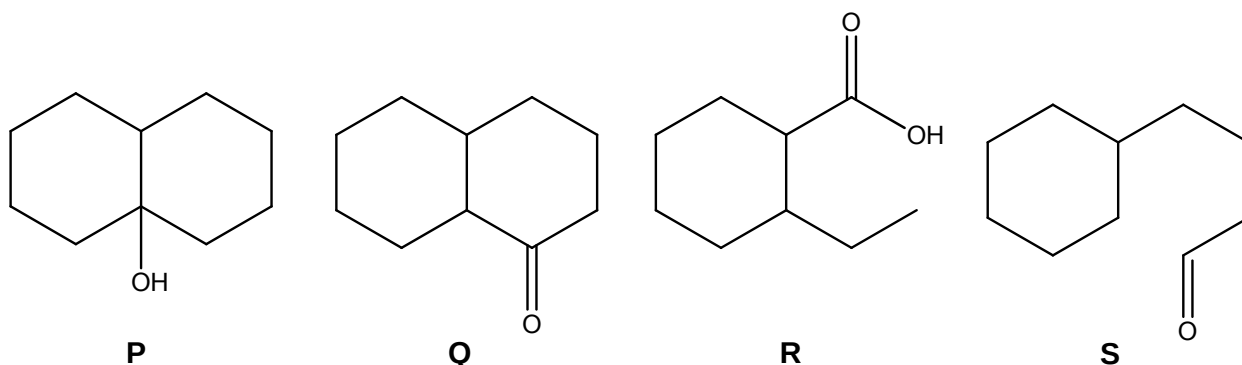
- A
- B
- C
- D

- 22 A sample of bromoethane was warmed with ethanolic silver nitrate, and cream AgBr precipitate was observed after about 4 minutes.

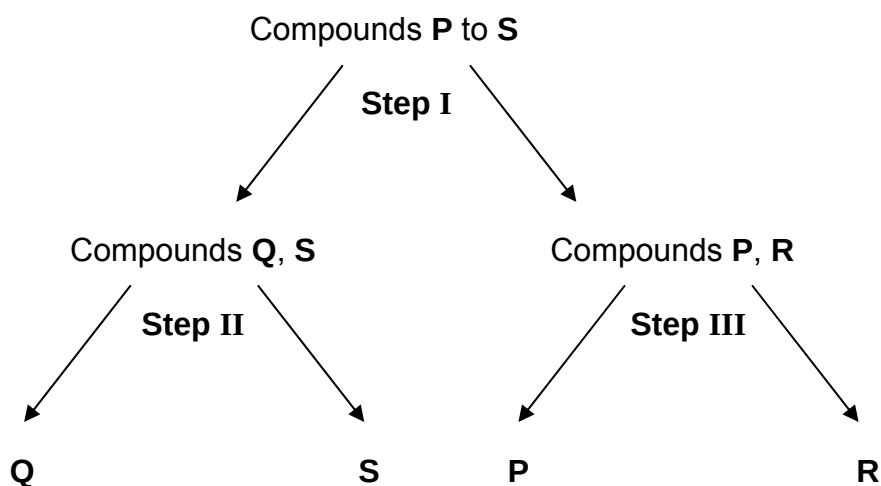
Under similar reaction conditions, which one of the following compounds will result in precipitate formation only after 8 minutes?

- | | | | |
|----------|------------------------|----------|--------------|
| A | iodoethane | B | chloroethane |
| C | 1-bromo-2-phenylethane | D | bromopropane |

- 23 Four unlabelled reagent bottles are known to contain the following compounds.



The laboratory assistant devised the following method to identify the 4 compounds.

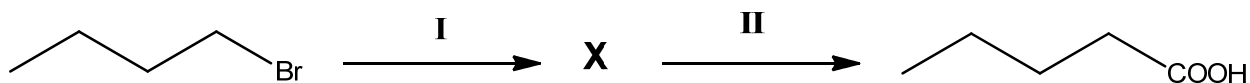


Which set of reagents and conditions was used in steps II and III respectively?

- | | Step II | Step III |
|----------|---|---|
| A | Addition of 2,4-DNPH | Addition of NaHCO ₃ |
| B | Reflux with acidified KMnO ₄ | Addition of NaHCO ₃ |
| C | Addition of 2,4-DNPH | Warm with acidified K ₂ Cr ₂ O ₇ |

D Reflux with acidified KMnO_4 Warm with acidified $\text{K}_2\text{Cr}_2\text{O}_7$

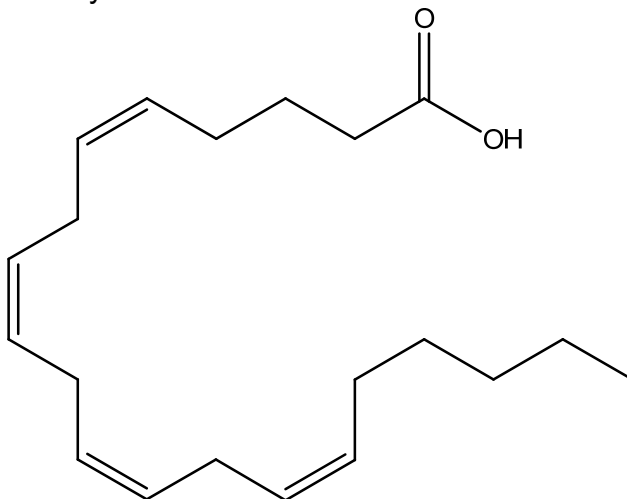
24 Carboxylic acids can be synthesized from halogenoalkanes in 2 steps:



Which one of the following statements about the reaction scheme is correct?

- A** X contains 3 π bonds.
- B** Hot aqueous sodium hydroxide can be used for step I.
- C** Hydrogen cyanide and trace amount of a base can be used for step I.
- D** Hot aqueous sulfuric acid can be used for step II.

25 Arachidonic acid is a polyunsaturated fatty acid that is involved in cell signaling in the body.



Which one of the statements about arachidonic acid is true?

- A** It reacts with hot acidic KMnO_4 to give effervescence.
- B** It does not dissolve in water and aqueous sodium hydroxide.
- C** It reacts with PCl_5 to give steamy fumes.
- D** It reacts with 2,4-dinitrophenylhydrazine to give an orange precipitate.

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements which you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

- 26** For which of the following pairs does the first species have a smaller bond angle?

- 1** XeF₄, SeF₂
2 CrO₄²⁻, SO₃²⁻
3 BH₃, SbH₃

- 27** Three substances, **X**, **Y** and **Z**, react to give a precipitate. The reaction mixture in a conical flask was placed on a piece of paper marked with a cross. The time taken, *t*, for various initial molar concentrations of **X**, **Y** and **Z** to produce enough precipitate to completely block the cross are given in the following table:

Experiment	[X]	[Y]	[Z]	<i>t</i> / s
1	0.5	0.05	0.05	120
2	2.0	0.05	0.05	120
3	1.0	0.10	0.05	30
4	1.0	0.10	0.10	15
5	1.5	0.05	0.10	?

Which statements follow from these results?

- 1** The rate equation for the reaction is rate = $k [\text{Y}]^2 [\text{Z}]$
2 The value of *t* of experiment 5 is 60 s.
3 The units of the rate constant are mol⁻¹ dm³ s⁻¹.

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

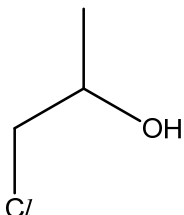
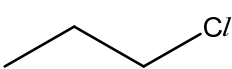
No other combination of statements is used as a correct response.

- 28** Element **W** is in the third period of the Periodic Table. The oxide of **W** has a giant ionic structure while the chloride of **W** has a simple molecular structure. Which of the following statements are correct?

- 1** The ionic radius of **W** is smaller than that of chlorine.
- 2** The second ionisation energy of **W** is lower than that of sodium.
- 3** The chloride of **W** can be used as a catalyst in the alkylation of benzene.

- 29** Propene is an important starting material in the petrochemical industry. It can yield a variety of chlorinated products, depending on the reagents and conditions applied.

Which of the following combinations are correct?

	Reagents	Major product
1	Aqueous Cl_2	
2	Acidified $\text{K}_2\text{Cr}_2\text{O}_7$	$\text{CH}_3\text{CO}_2\text{H}$
3	HCl(g)	

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

- 30** A species of ants found in South Africa secretes the compound **Z** below to warn the fellow ants from the same colony of dangers.



Z

Which of the following statements are true about compound **Z**?

- 1** **Z** exhibits geometric isomerism.
- 2** **Z** does not form an orange precipitate with 2,4 dinitrophenylhydrazine.
- 3** **Z** turns hot acidified potassium dichromate(VI) from orange to green.